A

rpoB

0.2

p__Pseudomonadota

p__Bdellovibrionota F

p__Bdellovibrionota

p_Thermodesulfobacteriota

p__Methylomirabilota

p__Campylobacterota

p__Dependentiae

p__Omnitrophota

p__Verrucomicrobiota

p__Chlamydiota

p__Sumerlaeota

p__JACRDZ01

p__Elusimicrobiota

p__Spirochaetota

p__UBA9089

p__Thermotogota

p__Fusobacteriota

p__Bacillota I

p__Margulisbacteria

p__Actinomycetota

p__Chloroflexota

p__Bacillota

p__UBP14

p__Krumholzibacteriota

p__Latescibacterota

p__Electryoneota

p__Bacteroidota

B

tef1α

0.05

p__Arthropoda

p__Arthropoda

p__Nematoda

p__Chaetognatha

p__Echinodermata

p__Mollusca

p__Platyhelminthes

p__Cnidaria

p__Mollusca

p__Cnidaria

p__Platyhelminthes

p__Arthropoda

p__Ascomycota

p__Apicomplexa

p__Ctenophora

p__Euglenozoa

p__Fornicata

p__Fornicata

p__Microsporidia

p__Bacillariophyta

C

Tissue-type

Brain

Gill

Kidney

Spleen

Intestine

Muscles

Liver

Skin

Astro-Poty

0.5

Mamastro-

Avastro-

Potyviridae

Quenya

0.5

Blubervirales

0.2

Avihepadna-

Herpetohepadna-

Orthohepadna-

Herpesvirales

0.2

Salmo-

Icta-

Ictaluri-

Cy-

Qinvirus

0.2

Ying-

Zurhausenvirales

0.5

Firstpapillomavirinae

Secondpapillomavirinae

Orthopolintovirales

0.5

Betadinto-

Alphadinto-

Ortervirales

0.2

Epsilonretro-

Gammaretro-

Simiispuma-

Betaretro-

Deltaretro-

Lenti-

Cysto

0.5

Flavi

0.5

Flavi-

Jingmen-

Pesti-

Pegi-

Hepaci-

Mono-Chu

0.5

Rhabdoviridae

Lispiviridae

Pneumoviridae

Filoviridae

Paramyxoviridae

Artoviridae

Bornaviridae

Chuviridae

Bunya-Arena

0.5

Peribunyaviridae

Fimoviridae

Cruliviridae

Tospoviridae

Hantaviridae

Phasmaviridae

Phenuiviridae

Orthonairovirus

Arenaviridaea

Nido

0.5

Coronaviridae

Mesoniviridae

Euroniviridae

Tobaniviridae

Arteriviridae

Pimascovirales

0.5

Iridoviridae

Marseilleviridae

Ascoviridae

Hepe-Virga

0.5

Virgaviridae

Bromoviridae

Closteroviridae

Togaviridae

Endornaviridae

Hepeviridae

Betaflexiviridae

Deltaflexiviridae

Alphaflexiviridae

Partiti-Picobirna

0.5

Partitiviridae

Deltapartiti-

Picobirnaviridae

Gammapartiti-

Orthomyxo

0.5

Quaranja-

Influenza-

Thogoto-

Amnoonviridae

Luteo-Sobemo

0.5

Alvernaviridae

Solemoviridae

Cirlivirales

0.5

Circoviridae

Vilyaviridae

Picorna

0.5

Caliciviridae

Polycipiviridae

Picornaviridae

Secoviridae

Iflaviridae

Dicistroviridae

Marnaviridae

Tombus-Noda

0.5

Tombusviridae

Nodaviridae

Piccovirales

0.5

Parvovirinae

Hamaparvovirinae

Densovirinae

Toti

0.5

Totiviridae

Toti-

Chrysoviridae

Totiviridae

Reo

0.5

Orbi-

Phytoreo-

Seadorna-

Orthoreo-

Aquareo-

Rota-

Cypo-

Dinoverna-

Oryza-

Colti-

Fiji-

Narna-Levi

0.5

Narnaviridae

Mitoviridae

Ourmia-

Leviviridae